

PILOT MOUND, MB, Canada

WMO: 711480

Lat: 49.1881N

Lon: 98.9064W

Elev: 1557

StdP: 13.89

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-20.0	-31.9	1.0	-23.5	-26.9	1.3	-19.2	29.7	9.7	27.5	10.2	7.3	290	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	84.6	69.8	81.4	68.1	78.6	65.9	73.7	80.7	71.0	78.5	68.7	75.8	12.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	123.3	78.5	68.5	110.9	75.6	66.1	101.8	73.1	38.4	80.9	36.0	78.5	33.9	75.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	23.5	21.0	DB	-30.1	91.5	4.4	3.0	-33.3	93.7	-35.9	95.4	-38.3	97.1	-41.5	99.3	(4)
(5)			WB	-30.7	77.6	5.0	2.9	-34.3	79.6	-37.2	81.3	-40.0	82.9	-43.6	85.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	4.3	7.5	20.8	37.8	50.7	60.7	64.9	63.6	55.1	40.6	23.9	9.7		
	DBStd	24.26	14.62	13.81	13.82	10.81	8.59	6.42	5.41	5.99	8.24	9.22	12.13	13.75		
	HDD50	6388	1418	1189	906	383	99	5	0	0	40	316	782	1250		
	HDD65	10467	1883	1609	1370	817	449	158	69	98	311	756	1232	1715		
	CDD50	1565	0	0	1	16	119	324	463	423	193	26	0	0		
	CDD65	170	0	0	0	0	5	27	67	56	14	1	0	0		
	CDH74	1958	0	0	0	3	108	297	631	676	226	17	0	0		
	CDH80	429	0	0	0	0	22	47	117	187	53	3	0	0		
	WSAvg	10.2	10.4	10.4	11.0	11.4	11.6	9.7	8.0	8.5	9.8	10.8	10.5	10.1		
	PrecAvg	19.5	0.6	0.5	0.7	1.0	2.9	3.6	3.0	2.7	1.7	1.3	0.7	0.7		
	PrecMax	27.9	1.2	1.6	2.4	2.2	6.3	7.6	6.5	6.9	4.6	5.6	2.4	1.9		
(15) Precipitation	PrecMin	12.6	0.0	0.0	0.0	0.0	1.3	1.2	0.7	0.7	0.1	0.1	0.0	0.0		
	PrecStd	3.9	0.3	0.4	0.6	0.6	1.3	1.4	1.3	1.5	0.9	1.1	0.7	0.5		
	0.4%	DB	38.6	40.2	58.1	72.5	82.8	85.8	87.0	90.3	86.3	75.7	57.5	41.1		
	MCWB	35.2	36.5	50.1	54.1	61.6	70.7	76.4	69.8	66.9	58.5	47.9	36.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	33.8	35.0	47.9	66.2	76.7	80.6	83.1	85.0	80.2	66.6	49.8	35.5		
	MCWB	31.9	32.8	42.2	51.2	56.6	66.8	72.5	69.2	63.7	53.8	42.5	33.0			
	5%	DB	29.9	32.2	41.3	61.2	71.7	77.5	80.3	81.1	75.7	60.6	44.0	31.3		
	MCWB	28.7	30.8	38.0	48.9	55.2	64.7	70.4	67.5	61.7	51.1	39.5	29.8			
	10%	DB	25.3	28.0	36.6	55.6	67.1	74.4	77.5	77.4	70.5	55.3	39.0	27.8		
	MCWB	24.3	26.7	34.5	46.3	53.7	63.0	68.2	65.2	59.0	48.5	36.1	26.7			
	0.4%	WB	35.2	37.4	50.6	57.6	64.8	75.8	79.2	75.5	70.3	60.2	49.1	36.9		
	MCDB	37.9	40.1	57.5	66.4	75.6	80.7	84.5	83.4	80.9	71.8	57.5	40.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.2	33.2	43.2	53.1	61.2	70.8	74.8	71.9	66.4	56.1	43.3	33.4		
	MCDB	33.7	34.7	48.2	63.5	69.4	77.3	80.9	81.3	75.4	65.2	48.1	35.8			
	5%	WB	28.8	31.0	38.3	49.9	58.4	67.2	72.1	69.4	63.5	52.5	39.3	30.2		
	MCDB	30.1	32.5	41.8	59.9	66.9	74.0	78.8	78.1	72.0	59.1	43.4	31.6			
	10%	WB	24.7	26.9	35.0	46.7	55.8	64.4	69.8	67.1	60.8	48.8	36.1	27.1		
	MCDB	25.6	28.3	37.0	55.4	64.5	71.8	76.1	74.8	69.6	55.1	38.8	28.2			
	5% DB	MDBR	16.5	17.6	16.5	19.9	22.3	19.7	20.8	23.2	22.8	18.6	15.9	15.4		
	MCDBR	19.8	18.4	20.1	30.2	31.0	25.0	24.6	28.6	30.6	29.2	23.8	18.8			
(36) Mean Daily Temperature Range	MCWBR	18.2	16.4	14.9	17.1	15.6	13.6	15.2	14.9	15.8	17.3	17.1	16.4			
	5% WB	MCDDBR	19.4	18.3	19.8	27.7	23.2	21.1	22.8	24.6	25.6	26.2	21.9	18.5		
	MCWBR	18.0	16.2	14.9	16.8	13.8	13.8	15.5	14.5	15.2	16.3	16.3	16.3			
(40) Clear-Sky Solar Irradiance	taub	0.267	0.276	0.290	0.352	0.366	0.372	0.379	0.378	0.345	0.293	0.264	0.244			
	taud	2.275	2.273	2.411	2.326	2.343	2.366	2.353	2.346	2.424	2.548	2.486	2.402			
	Ebn at Noon	251	278	295	284	283	281	278	272	272	273	257	250			
	Edh at Noon	23	29	30	37	38	37	37	36	30	22	18	17			
(44) All-Sky Solar Radiation	RadAvg	481	853	1248	1583	1768	1905	2022	1662	1216	749	428	341			
	RadStd	37	70	102	115	131	127	90	110	88	95	59	41			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.12	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page